

Investigating syntactic effects in NPI illusions in Turkish

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Background

Negative Polarity Items (NPIs) such as ever are licensed when they are within the scope of a negative element such as **no**:

(1) **No** bills [that the senators voted for] will ever become law.

BUT there is an illusion of acceptability with an intrusive licenser [1-3]:

(2) *The bills [that **no** senator voted for] will ever become law.

Languages with **prospective NPI licensing**, i.e., negation follows the NPI, such as Turkish [4], Japanese [5], Korean [6] showed illusion with intrusive licensers in complement clauses in offline data.

(3) *Kimse [Ali-nin çalış-ma-diğ-in-ı] söyle-di.
Anybody Ali-GEN work-NEG-FN-POSS-ACC say-PST
“Anybody said that Ali didn’t work.”

? *Semantic accounts* [1] attribute illusion to the indirect licensing by the negative inferences of relative clauses, but illusion is NOT predicted with complement clauses, which do not lead to such negative inferences.

? *Cue-based memory accounts* [7] attribute illusion to partially matching cues of the intrusive licenser and the NPI in memory retrieval processes, but illusion in prospective NPI licensing can NOT be explained with retrieval as the licensing negation follows the NPI.

The present study:

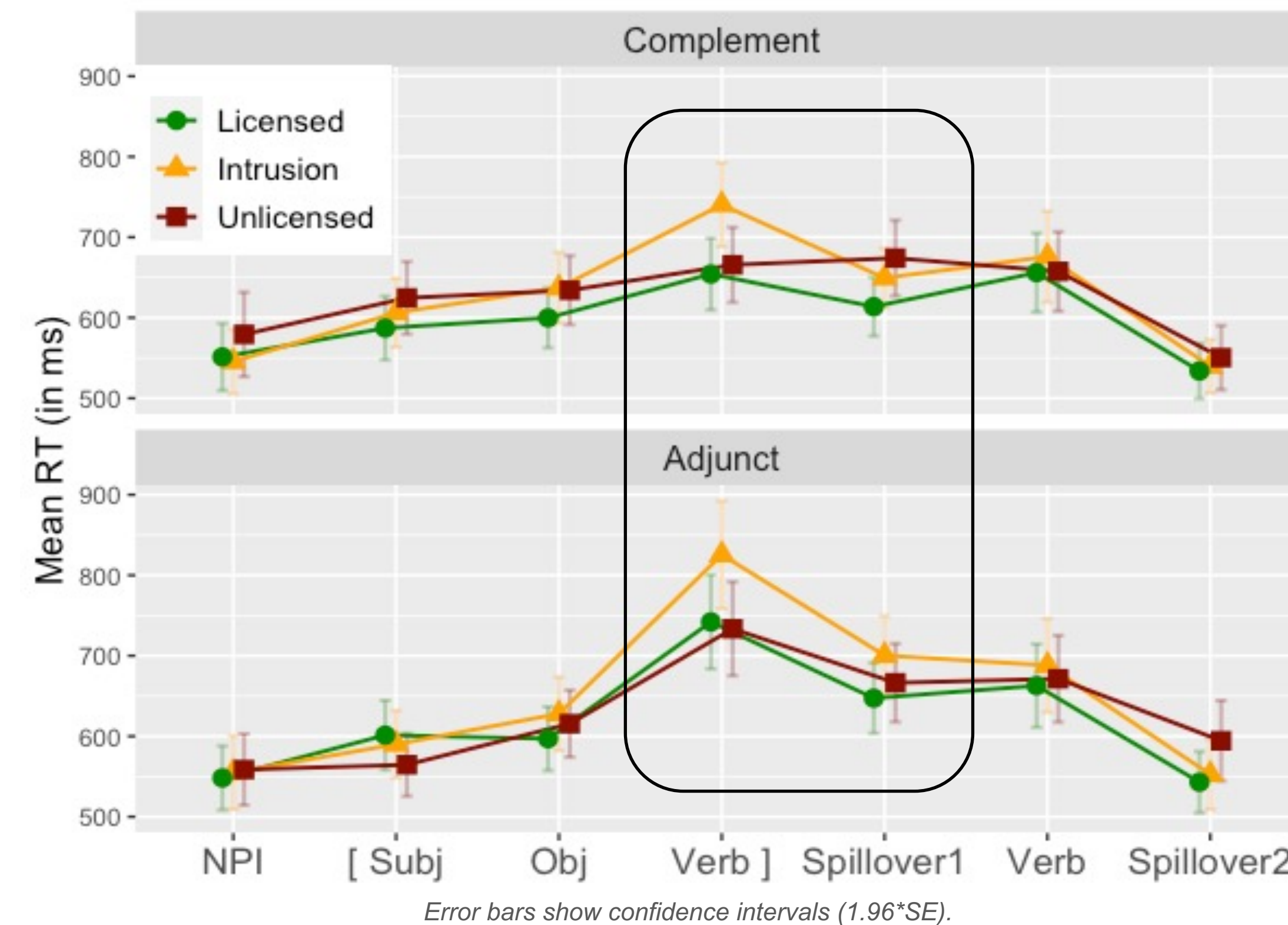
Q1: Can the NPI illusion observed in Turkish [5] be replicated? *Yes!*

Q2: Does the illusion differ depending on the syntactic position of the inappropriate licenser, i.e., complement vs. adjunct clauses? *No!*

Self-Paced Reading Task

($N_{\text{subject}} = 55$, $N_{\text{item}} = 24$, $N_{\text{filler}} = 48$)

Is there online evidence for NPI illusion in Turkish [cf.5]? ✓



EmbeddedVerb:

Intrusion (V-neg) vs. Licensed & Unlicensed (V) (intrusion)

Main verb: (helmert contrast)

Licensed (V-neg) vs. Intrusion & Unlicensed (V) (grammaticality)
Intrusion (V-neg...V) vs. Unlicensed (V...V) (intrusion)

Intrusion effect (β 's $\geq .04$, t 's ≥ 2.43 , p 's $\leq .05$)

Embedded verb & Spillover1

Clause effect ($\beta = .08$, $t = 3.84$, $p < .001$)

Embedded verb: Adjunct > Complement

No effects (β 's $\leq .03$, t 's ≤ 1.39 , p 's $\geq .17$)

Main verb & Spillover2

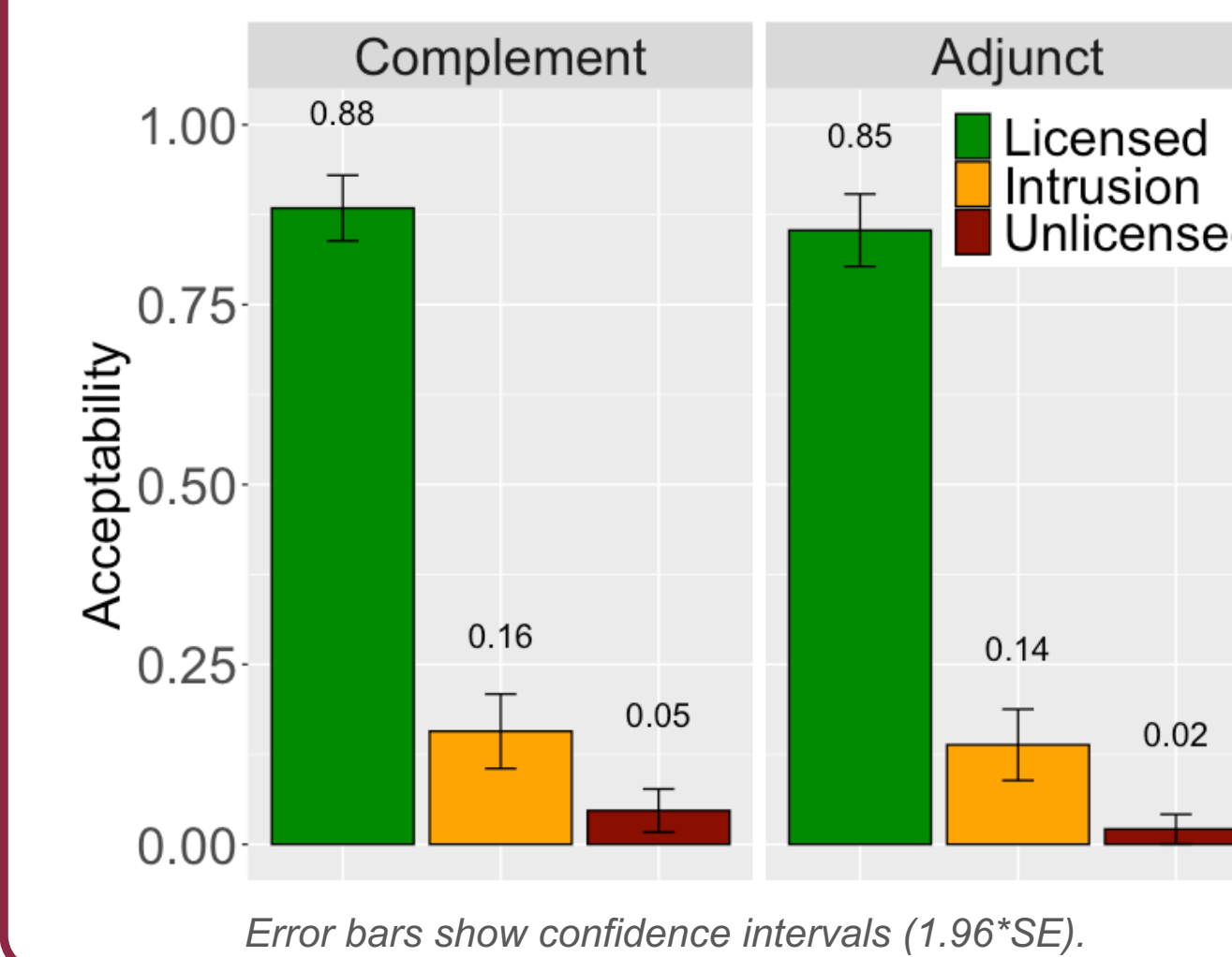
No Interactions (β 's $\leq .06$, t 's ≤ 1.50 , p 's $\geq .13$)

Speeded Acceptability Judgment Task

($N_{\text{subject}} = 48$, $N_{\text{item}} = 24$, $N_{\text{filler}} = 48$)

- Can the illusion with complement clauses be replicated [4] and extended to adjunct clauses? ✓

- Does the illusion differ depending on the syntactic position of the intrusive licenser? ✗



Grammaticality effect

($\beta = 4.99$, $z = 17.55$, $p < .001$)

Intrusion effect

($\beta = 1.67$, $z = 5.08$, $p < .001$)

No Clause effect

No interactions

(β 's $\geq .68$, z 's $\leq .98$, p 's $\geq .33$)

Discussion

→ The NPI illusion in Turkish in the offline study **replicates** earlier findings with complement clauses [4], shows that illusion appears with **adjunct** clauses as well.

→ Overall easier processing in complements than adjuncts shows **delayed structure building** in adjuncts [9].

→ No facilitation in intrusion aligns with earlier findings in other languages with prospective NPI licensing [5-6, cf. 2, 4].

→ **Inhibition in intrusion** [6] may be due to the cost of updating the probabilities of sequences to have a larger shift towards nonveridical ones that allow for a dependency between embedded verb and matrix NPI [10].

(5a) [NPI secretary's office_{obj} call-**neg**-...] ...

(5b) Secretary [NPI office-ACC call-**neg**-...] ...

Future direction to test (5b): #The mirror [anybody ... didn't shatter] ...

(6) *Kimse [ayna-nın hala parçalan-ma-diğ-in-a] şaşır-dı.
“Anybody was surprised that the mirror didn’t shatter yet.”

! Inhibition is *unlikely* to be due to negation only! Ask why.

→ Absence of slow-down in unlicensed conditions may be due to the parser's ongoing expectations for a licenser [6]. A **maze-task** may be more suited to reveal such effects [11].

Experimental Stimuli & Methodology

- 2 x 3 within-subjects design, crossing
 - Clause Type: Complement or Adjunct Clause
 - Negation: Negated Matrix Verb (Licensed), Negated Embedded Verb (Intrusion) or No Negation (Unlicensed)
- NPI kimse as the matrix subject
- Embedded subject as a generic or proper noun, e.g. *Ali(-nin)*
- Words in italic, i.e., *şüphesiz* ‘no doubt’ and *zaten* ‘anyways’, added as spillover regions in the self-paced reading task
- Data analyzed with logistic/linear regression models [8]

NPI [Subject ... VP(-neg)] ... VP(-neg)

(4) **Complement** Kimse [sekreter-ler-in ofis-i ara-(**ma**)-diğ-in-a] *şüphesiz* kız-(**ma**)-dı *zaten*.
Anybody secretary-PL-GEN office-ACC call-(NEG-)FN-POSS-DAT no.doubt be.mad-(NEG-)PST anyway

Adjunct Kimse [sekreter-ler-ø ofis-i ara-(**ma**)-diğ-in-da] *şüphesiz* kız-(**ma**)-dı *zaten*.
Anybody secretary-PL-NOM office-ACC call-(NEG-)FN-POSS-LOC no.doubt be.mad-(NEG-)PST anyway
“Anybody was mad *that/when* the secretaries called/didn’t call the office.”

References. [1] Xiang et al., 2009. [2] Parker & Phillips, 2016. [3] Drenhaus et al., 2005. [4] Yanilmaz & Drury, 2017. [5] Su & Aparicio, 2022. [6] Lee & Yun, 2022. [7] Vasishth et al., 2009. [8] Bates et al., 2015. [9] Frazier & Clifton, 1996. [10] Levy, 2008. [11] Boyce et al. 2020.
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